

A physics-assisted deep learning method for high-fidelity breast microwave image reconstruction in electromagnetic inverse scattering

Cite as: J. Appl. Phys. **139**, 184903 (2026); doi: [10.1063/5.0326725](https://doi.org/10.1063/5.0326725)

Submitted: 7 February 2026 · Accepted: 23 April 2026 ·

Published Online: 11 May 2026



Min Lu,^{1,a)} Chen Wang,¹ Xia Xiao,² Lei Yuan,¹ and Yuan Yang¹

AFFILIATIONS

¹School of Automation and Information Engineering, Xi'an University of Technology, Xi'an 710048, China

²Tianjin Key Laboratory of Imaging and Sensing Microelectronic Technology, School of Microelectronics, Tianjin University, Tianjin 300072, China

^{a)}Author to whom correspondence should be addressed: minlu@xaut.edu.cn

ABSTRACT

Microwave imaging is a promising next-generation breast imaging technology. At present, cutting-edge microwave image reconstruction methods based on deep learning (such as various variants of convolutional neural networks) mostly adopt the general convolution modules and architectures from visual image processing and do not fully integrate physical priors. For breast tissues with strong heterogeneity and complex structures, the existing methods have prominent drawbacks, such as limited reconstruction accuracy and excessive model parameters. To address these issues, this paper proposes a lightweight deep learning framework based on physical assistance for high-fidelity microwave image reconstruction. First, a physically reliable initial reconstruction is generated using the Born iterative method. Subsequently, the Microwave Image Enhancement Network (MIENet) is proposed to reconstruct high-fidelity microwave images with rich details. Within the MIENet architecture, a group progressive residual module is proposed that achieves both superior detail reconstruction capability and computational efficiency, and the convolutional block attention module is integrated into the U-shaped architecture to enable the selective learning of key features. In addition, a hybrid loss function integrating multiple evaluation criteria is proposed, which effectively guides the network to generate high-fidelity microwave images. Experiments conducted on the breast dataset show that MIENet outperforms classical image reconstruction architectures in terms of image quality while being more lightweight: it achieves a peak signal-to-noise ratio of 39.14 dB and a structural similarity index of 0.9904, with a total parameter count and floating-point operations accounting for only 67.16% and 1.3% of those of state-of-the-art networks, respectively.

Published under an exclusive license by AIP Publishing. <https://doi.org/10.1063/5.0326725>

I. INTRODUCTION

Microwave imaging technology enables the visualization of scenes that are not directly accessible or visible. Owing to its inherent advantages, such as prospects for miniaturization and non-contact operation, it has been extensively studied across various fields, including through-the-wall imaging¹ and breast imaging,² and so on. By solving the corresponding inverse scattering problem, the dielectric properties inside an unknown object can be reconstructed to achieve visualization of its internal structure.³

Inverse scattering imaging is an ill-posed and nonlinear problem. Solution approaches can be broadly categorized into two

classes. One is the approximation method that converts nonlinear problems into linear problems, such as backpropagation and Born approximation.⁴ The other approach is to construct objective functions related to nonlinearity, which are then solved iteratively for image reconstruction, such as Born Iterative Method (BIM),⁵ Distorted Born Iterative Method (DBIM),⁶ Contrast Source Inversion (CSI),⁷ and subspace-based optimization method.⁸ In the field of microwave breast imaging, the high dielectric contrast between adipose and fibroglandular tissues poses a significant challenge to breast microwave imaging.⁹ Consequently, various improved Born methods were proposed to overcome this strong nonlinear problem. A multistage algorithm combining prior eigenfunctions was utilized to improve the stability of breast image

reconstruction.¹⁰ Batista *et al.*³ proposed multi-frequency strategies and regularization methods to reduce the number of iterations in solving the inverse problem. Given the relatively small changes in electromagnetic (EM) characteristics within similar tissues, structural prior knowledge derived from MRI images was introduced into inverse scatter imaging, which effectively improved the imaging quality.¹¹ Sato and Kidera¹² proposed the CSI method with region of interest limitation to reduce the number of unknown solutions and enhance the dielectric profile imaging of the breast. A more detailed and comprehensive review on electromagnetic imaging can be found in Ref. 13. Although traditional inverse scattering imaging has improved the accuracy of solutions to a certain extent, it remains prone to getting stuck in local minima and yields limited imaging quality.

With the rapid advancement and powerful performance of deep learning (DL) in tasks, such as target localization,¹⁴ image prediction,¹⁵ and image enhancement,¹⁶ DL-based inverse scattering imaging methods have garnered substantial attention. The most straightforward implementation scheme is to take the measured field data as input and the reconstructed electromagnetic image as output. In this paradigm, the deep learning network acts as a black box to fit the inverse scattering imaging process. Ambrosiano *et al.*¹⁷ fed the scattering matrix into a fully-connected network to obtain breast dielectric images, achieving results superior to those of the CSI method. Qin *et al.*¹⁸ employed a convolutional neural network (CNN) combined with ultrasound and microwave field data to image the breast. The most classic neural network in image reconstruction is the encoder-decoder network. This type of network¹⁹ and its variants^{20,21} are commonly used to perform inverse scattering imaging of inhomogeneous targets (e.g., handwritten digits) by feeding scattering field data as input. Zhang *et al.*²² proposed FPNNet + ResU-Net, which takes time-domain acoustic signals as input to generate thermoacoustic tomography images. In addition, several studies incorporated iterative updates of variables from physical models into the learning process of neural networks.^{23,24}

Wei and Chen²⁵ compared three training schemes by feeding the UNet network with: (1) raw scattering field data; (2) contrast images obtained by backpropagation (BP) (note that it is not the BP algorithm used for weight updates in deep learning) algorithm; and (3) dominant component of induced current. The findings indicated that the latter two physics-assisted approaches outperformed direct learning from scattering data, as the networks avoid unnecessary costs to learn potentially complex physical knowledge. By incorporating preliminary results derived from physical principles, the network becomes more tractable to train and achieves better performance.

By taking the initial image from inverse scattering imaging as the network input and then applying deep learning architectures to enhance image quality, the problem is transformed into an image-to-image mapping task. Li *et al.*²⁶ were the first to use deep neural networks (DNNs) to solve electromagnetic inverse scattering problems, employing backpropagation (BP) images as input to the DNN and achieving significantly higher image quality for handwritten digit reconstruction compared with BP and CSI methods. Shah and Moghaddam²⁷ adopted a super-resolution convolutional neural network (SRCNN) to improve the spatial

resolution of microwave breast imaging. A 3D CNN²⁸ was proposed to reconstruct microwave breast images from images containing artifacts. In this image-to-image reconstruction mode, the most typical network is still the encoder-decoder network. Huang *et al.*²⁹ used the structural similarity index (SSIM) loss function within UNet to achieve image reconstruction of digital handwriting, further improving image quality. Ruiz *et al.*³⁰ acquired qualitative images through orthogonal sampling and fed them into a UNet, achieving favorable results in imaging piecewise homogeneous targets. Hirose *et al.*³¹ proposed a deep convolutional autoencoder scheme based on the CSI algorithm to enhance the reconstruction accuracy of microwave breast images. Maricar *et al.*³² proposed an Attention UNet architecture with Attention Gates for microwave imaging of simple-shaped targets, which achieved performance improvements compared to other CNN architectures. Other studies^{33,34} incorporated prior knowledge from ultrasound to improve the breast reconstruction effect using UNet, but the image quality is still relatively limited. These studies demonstrate the superior performance and potential of machine learning methods in solving the inverse scattering problem, actively promoting the progress of high-quality electromagnetic imaging. Existing microwave imaging architectures often rely on generic simple convolution modules and networks in the field of natural visual images. However, natural visual images and microwave images based on physical reconstruction have different image forms and meanings. These general architectures exhibit stable performance in simple geometric shape tasks, but they often show lower accuracy when applied to complex reconstruction scenarios involving highly heterogeneous and structurally complex breast tissue. Therefore, the development of specialized modules and the optimization of network structures for heterogeneous breast imaging are crucial for enhancing feature extraction and high-quality reconstruction of complex breast tissues.

To address the existing deficiencies and advance the research on microwave imaging, this paper proposes a physics-assisted microwave image reconstruction framework, which can be applied to complex breast imaging. The main advantages and contributions of this paper are as follows:

- (1) A Microwave Image Enhancement Network (MIENet) is proposed, which utilizes initial imaging from the BIM method to achieve high-fidelity microwave image reconstruction. This architecture combines Group Progressive Residual Module (GPRM), Convolutional Block Attention Module (CBAM) and a hybrid loss function, which can selectively focus on, extract, and efficiently propagate key features in images, effectively improving the performance of microwave image reconstruction for complex structures.
- (2) The GPRM is proposed for the process of image encoding-decoding. Its progressive grouping strategy ensures the extraction and efficient combination of rich details to form high-quality microwave images. The residual connection mechanism in the module is incorporated to alleviate feature degradation in deep networks and ensure complete transmission of critical information.
- (3) A multi-criteria hybrid loss function is designed, which comprises L_1 loss, SSIM loss, edge loss, and perceptual loss. The

loss function evaluates both pixel-level accuracy and overall image error while incorporating high-level semantic information from a pre-trained segmentation network into the optimization process of the enhancement network, thereby guiding the network to converge more stably toward the global optimal solution.

- (4) Comprehensive performance evaluations are conducted on microwave breast data. Comparisons are implemented with various advanced deep learning networks, and performance and efficiency are thoroughly analyzed. Furthermore, the contributions of each module in the proposed method are discussed through ablation experiments. The complexity of the proposed network is also evaluated.

The remainder of this paper is organized as follows. Section II provides the theoretical background of the inverse scattering problem. Section III presents the formulation of the inverse algorithm, the proposed MIENet, and detailed descriptions of its key modules. Section IV introduces the experimental implementation details and evaluation metrics. Section V reports the evaluation results and discussions, including comparisons with various advanced architectures and ablation experiment results. Finally, Sec. VI provides the conclusions and prospects of this work.

II. THEORETICAL BACKGROUND OF THE INVERSE SCATTERING PROBLEM

Consider a dielectric scattering problem in a homogeneous background medium, where transmitters and receivers illuminate the scatterer and receive the scattered signals, respectively, as illustrated in Fig. 1. In the following, only non-magnetic media will be considered and magnetic current sources will be neglected. The total electric field E^{tot} within region D can be decomposed into two parts: the incident field E^{inc} in the absence of the scatterer, and the scatter field E^s generated by the interaction between the source and the scatterer when the scatterer is present, that is, $E^{tot}(\mathbf{r}) = E^{inc}(\mathbf{r}) + E^s(\mathbf{r})$. Considering the scattering problem under a 2D transverse magnetic (TM) wave incidence, where the scatterer with relative permittivity $\epsilon_r(\mathbf{r})$ is located in the 2D region D . The contrast $\chi(\mathbf{r})$ is defined as $\epsilon_r(\mathbf{r}) - 1$, and the induced contrast current density is $J(\mathbf{r}) = (\epsilon_r(\mathbf{r}) - 1)E^{tot}(\mathbf{r})$. According to the Lippmann-Schwinger equation, the electric field integral equation can be expressed as²³

$$E^{tot}(\mathbf{r}) = E^{inc}(\mathbf{r}) + k_0^2 \int_D g(\mathbf{r}, \mathbf{r}') J(\mathbf{r}') d\mathbf{r}', \mathbf{r} \in D, \quad (1)$$

where $g(\mathbf{r}, \mathbf{r}')$ represents the scalar Green's function. Parameter $k_0 = \omega \sqrt{\mu_0 \epsilon_0}$ is free space wavenumber. Parameters ω , μ_0 , and ϵ_0 denote angular frequency, magnetic permeability, and dielectric constant in free space, respectively. The scattering field integral equation is

$$E^s(\mathbf{r}) = k_0^2 \int_D g(\mathbf{r}, \mathbf{r}') J(\mathbf{r}') d\mathbf{r}', \mathbf{r} \in S. \quad (2)$$

It is difficult to directly solve the integral equation to obtain an analytical solution, so approximate solutions are often obtained

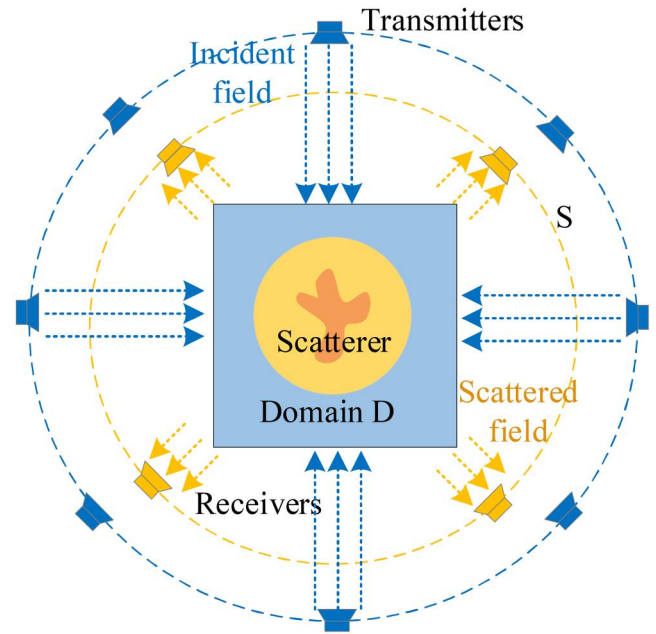


FIG. 1. Schematic diagram of the electromagnetic (EM) inverse scattering problem.

through discretization methods. First, the antenna located at $\mathbf{r}_p^i$ emits $N_i (p = 1, 2, \dots, N_i)$ times, and the scattering electric field values are measured by N_s antennas at positions $\mathbf{r}_q^s (q = 1, 2, \dots, N_s)$. The solution region D is discretized into M square grids. The discretized current source is represented by the following matrix equation:

$$\bar{J} = \bar{\chi} \cdot \bar{E}^{tot} = \bar{\chi} \cdot (\bar{E}^{inc} + \bar{G}_D \cdot \bar{J}). \quad (3)$$

Among them, $\bar{E}^{inc} = [E^{inc}(\mathbf{r}_1), \dots, E^{inc}(\mathbf{r}_M)]^T$ and $\bar{J} = [J(\mathbf{r}_1), \dots, J(\mathbf{r}_M)]^T$ are vectors, and $\bar{\chi}$ represents the matrix containing $\chi(\mathbf{r}_m)$. $\bar{G}_D$ is a $M \times M$ matrix. The scattering field $\bar{E}^s$ at the receiver can be represented as

$$\bar{E}^s = \bar{G}_S \cdot \bar{J}, \quad (4)$$

where $\bar{G}_S$ denotes the radiation operator of size $N_s \times M$.¹³

III. PHYSICS- ASSISTED DEEP LEARNING FOR MICROWAVE IMAGING

The overall workflow of the proposed physics-assisted microwave imaging approach is illustrated in Fig. 2. The scattered field data are processed using the physically reliable BIM method to obtain a preliminary reconstruction, which is a physical mapping from the field data to spatial pixels. Then, the MIENet is proposed for microwave image enhancement, enabling high-fidelity recovery of image contours and fine details.

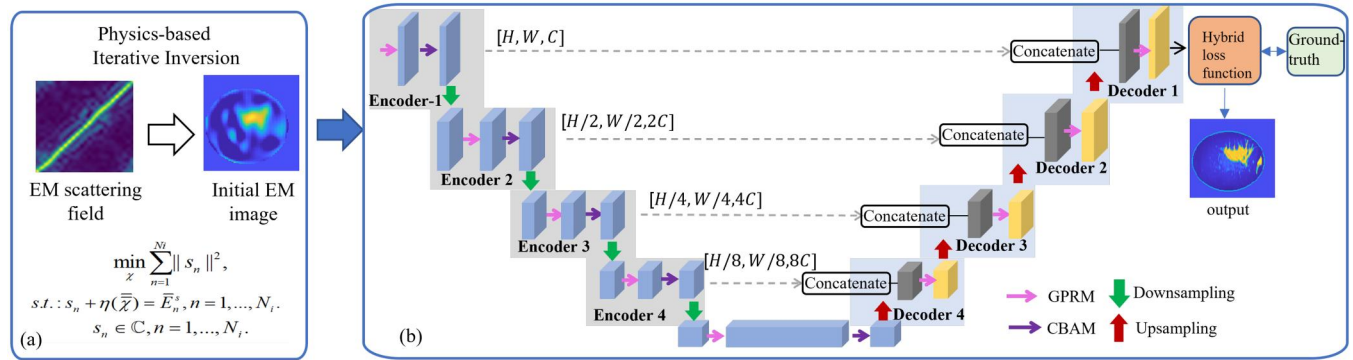


FIG. 2. Flowchart of the proposed framework. (a) Initial physical reconstruction. (b) MIENet architecture.

A. Inversion algorithm

The collected electromagnetic signals are initially reconstructed into a spatial image through the inverse algorithm. The key of the inverse scattering problem is to solve the contrast $\bar{\chi}$, through which the dielectric distribution of the scatterer can be obtained. By Eqs. (3) and (4), the scattering field can be expressed as

$$\bar{E}^S = \bar{G}_S \cdot (\bar{I} - \bar{\chi} \cdot \bar{G}_D)^{-1} \cdot (\bar{\chi} \cdot \bar{E}^{inc}), \quad (5)$$

where $\bar{I}$ represents the identity matrix, and this equation can be simplified and described using an operator $\eta(\cdot)$,

$$\bar{E}^S = \eta(\bar{\chi}). \quad (6)$$

BIM¹³ employs an iterative algorithm to gradually improve the reconstructed electromagnetic distribution of scatterers. By selecting an initial value χ_0 , the scattered field is calculated based on the current estimate (forward problem), and the residual norm of computed and measured scattered fields is taken as the objective function. By setting the maximum number of iterations, the final reconstruction result can be obtained. The contrast function can be solved as follows:

$$\min_{\chi} \sum_{n=1}^{N_i} \|\bar{E}_n^S - \eta(\bar{\chi})\|^2. \quad (7)$$

Here, the slack variables are introduced for the convenience of numerical computation. The minimization of the sum of errors for each equation can also be expressed as the minimization of the slack variable s_n for each equation,³

$$\min_{\chi} \sum_{n=1}^{N_i} \|s_n\|^2, \quad (8)$$

$$s.t.: s_n + \eta(\bar{\chi}) = \bar{E}_n^S, n = 1, \dots, N_i. \quad (9)$$

$$s_n \in \mathbb{C}, n = 1, \dots, N_i. \quad (10)$$

When the scattering field reaches its optimal solution, the slack variables will be close to zero. In order to stabilize the solving process and control the physical rationality of the solution, the Tikhonov regularization term $T(\cdot)$ is added, the objective function is written as

$$\min_{s, \chi} \sum_{n=1}^{N_i} \|s_n\|^2 + \alpha T(\bar{\chi}), \quad (11)$$

where α is a regularization weight parameter used to optimize the update speed and stability of parameters. This description of solving the inverse problem is formulated as a model with a quadratic objective function and linear constraints, also known as a quadratic programming problem,³ and then it can be solved through the barrier method.³⁵

B. Deep learning architecture

This section proposes the Microwave Imaging Enhancement Network (MIENet), as shown in Fig. 2(b). Inspired by the superior performance of the UNet architecture in tasks such as image denoising and super-resolution reconstruction,³⁶ this paper adopts a typical U-shaped design for microwave image enhancement. The network features a symmetric encoder-decoder architecture. The encoder progressively extracts high-level semantic features of the input image through convolution and downsampling, compressing the spatial dimensions. Correspondingly, the decoder gradually recovers spatial resolution through upsampling, mapping high-level features to pixel-level details. Skip connections in the U-shaped network effectively mitigate information loss and the gradient vanishing problem. The number of feature channels is doubled at a fixed ratio in the downsampling process, while the feature size is halved accordingly. The architecture is flexible and can be configured to different depths as needed. The key modules employed in constructing MIENet are described below.

1. Group progressive residual module

The GPRM is proposed to improve the quality and efficiency of feature extraction and image restoration for microwave images

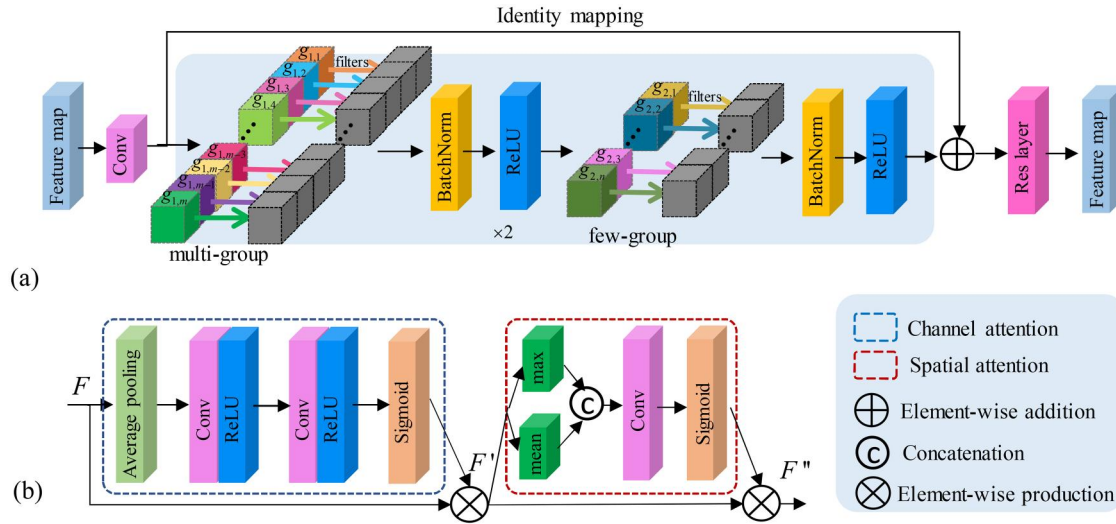


FIG. 3. (a) Proposed group progressive residual module. (b) Convolutional block attention module.

during the encoding and decoding processes of the U-shaped network, as illustrated in Fig. 3(a). The module begins by grouping convolutional channels and filters to perform independent convolutional operations within each group, after which the outputs are concatenated. Since details in different scattering regions of microwave images are often dominated by distinct feature patterns, and details in the same region may contain multi-scale and multi-dimensional information, a convolution strategy with a larger number of groups is adopted in the early part of the module. The convolutional channels are divided into multiple groups $\{g_{1,1}, g_{1,2}, \dots, g_{1,m}\}$ to increase the diversity of the feature space. This division allows dedicated learning channels to be allocated to different types of details as much as possible, thereby enhancing the network's expressive capability.

In the latter part of the module, a convolution strategy with fewer groups is adopted, with channels divided into fewer groups $\{g_{2,1}, g_{2,2}, \dots, g_{2,n}\}$, $g_{2,n} < g_{1,m}$. This enables the rich features extracted by the multi-group convolutions to achieve global correlations and interaction of detailed features through subsequent fewer-group convolutions. This progressive grouping strategy ensures that abundant details are extracted and effectively combined, contributing to the formation of high-quality microwave images rich in detailed information. In this paper, the number of multiple groups and few groups are set to 16 and 4, respectively. Furthermore, to mitigate the gradient vanishing problem in deep networks, residual connections are introduced into the module. The residual output $R(x)$ is given by

$$R(x) = f(x) + x, \quad (12)$$

where x is the feature input to the residual block, and $f(x)$ is the feature learned by the internal network layer of the residual block. By skipping the internal depth-wise connections within the

module and directly transferring shallow input features to deeper layers, the detailed features are prevented from being overwritten during multiple convolutions.

In this module, each grouped convolution is followed by Batch Normalization (BN) and a ReLU activation function. The BN layer can effectively solve the problem of covariate offset inside the network by normalizing the data in each batch, as follows:

$$y_k = BN_{\gamma, \beta}(x_k) = \gamma \hat{x}_k + \beta, \quad (13)$$

$$\hat{x}_k = \frac{(x_k - \mu_B)}{\sqrt{\delta_B^2 + c}}, \quad (14)$$

where γ and β are scaling and translation parameters, c is a constant, and parameters μ_B and δ_B^2 represent the mean and variance of each batch, respectively. The ReLU (Rectified Linear Unit) function is expressed as follows:

$$\text{ReLU}(x) = \max\{0, x\}. \quad (15)$$

The BN and ReLU layer can ensure the stability of input data distribution and improve the performance and training efficiency of neural networks.

2. Convolutional block attention module

The CBAM is incorporated into the proposed MIENet architecture, as illustrated in Fig. 3(b). When dealing with images of complex structures, it enables the network to selectively focus on and process key feature information while ignoring other irrelevant information. The CBAM, which combines Channel Attention Module (CAM) and Spatial Attention Module (SAM), can effectively improve the feature extraction performance of convolutional

layers. The expression is as follows:

$$F' = M_c(F) \otimes F, \quad (16)$$

$$F'' = M_s(F') \otimes F', \quad (17)$$

where F is the input feature map. The channel-wise and spatial-wise attention maps are obtained as follows:

$$M_c(F) = \sigma(\text{MLP}(\text{AvgPool}(F))), \quad (18)$$

$$M_s(F) = \sigma(\text{Conv}(\text{AvgPool}(F); \text{MaxPool}(F))), \quad (19)$$

where parameter σ denotes a sigmoid function. The average pooling in spatial dimension is used in the CAM. Multi-Layer Perceptron (MLP) consists of two 1×1 convolutional layers, as CAM focuses solely on the global channel information across the entire feature map. In SAM, average pooling and maximum pooling operations are performed along the channel axis. Their concatenated feature is then processed through a convolutional layer, followed by a sigmoid function to generate the output.

3. Hybrid loss function

The loss function is used to quantify the discrepancy between the predicted outputs and the ground-truth, thereby guiding the parameter updates of the network. It is crucial for the successful construction of deep learning networks.

To comprehensively incorporate multi-scale image information and mitigate the limitations of a single loss function, this study proposes a hybrid loss function that includes L_1 loss, SSIM loss, edge loss, and perceptual loss. This multi-criteria mechanism enables the model to evaluate reconstruction discrepancies from multiple perspectives, including pixel-level fidelity, overall structural similarity, edge preservation, and high-level semantic content. The hybrid loss function is defined as follows:

$$\text{Loss} = 0.25(L_1 + L_{\text{SSIM}} + L_{\text{edge}} + L_{\text{perceptual}}), \quad (20)$$

wherein the L_1 loss function calculates the mean absolute error between the enhanced image and the ground-truth image at the pixel level, and its expression is given as follows:

$$L_1 = \frac{1}{n} \sum_{i=1}^n |\varepsilon_r - \hat{\varepsilon}_r|, \quad (21)$$

where ε_r and $\hat{\varepsilon}_r$ denote ground-truth and reconstructed value of the i th sample, respectively, and n is the total number of samples.

SSIM loss (L_{SSIM}) is used to evaluate the similarity between images by jointly considering brightness, contrast, and structural factors. A larger SSIM value indicates less distortion between the images. The calculation formula is as follows:

$$\text{SSIM}(\varepsilon_r, \hat{\varepsilon}_r) = \frac{1}{J} \sum_{j=1}^J \frac{(2\mu_{\varepsilon_r}^j \mu_{\hat{\varepsilon}_r}^j + c_1)(2\sigma_{\varepsilon_r \hat{\varepsilon}_r}^j + c_2)}{(\mu_{\varepsilon_r}^2 + \mu_{\hat{\varepsilon}_r}^2 + c_1)(\sigma_{\varepsilon_r}^2 + \sigma_{\hat{\varepsilon}_r}^2 + c_2)}, \quad (22)$$

$$L_{\text{SSIM}} = 1 - \text{SSIM}(\varepsilon_r, \hat{\varepsilon}_r). \quad (23)$$

Here, $\mu_{\varepsilon_r}^j$ and $\mu_{\hat{\varepsilon}_r}^j$ are the mean value of the j th local window in images ε_r and $\hat{\varepsilon}_r$, respectively. $\sigma_{\varepsilon_r}^2$ and $\sigma_{\hat{\varepsilon}_r}^2$ are local variances of the j th local window in images ε_r and $\hat{\varepsilon}_r$. Parameter $\sigma_{\varepsilon_r \hat{\varepsilon}_r}^j$ is the covariance of local pixels in the two images. The constants c_1 and c_2 are used to ensure numerical stability. A smaller L_{SSIM} value indicates higher structural similarity between the images.

The edge loss (L_{edge}) is a loss function based on image edges, which detects edges through the Laplacian operator and calculates the average absolute error between these edge features,

$$L_{\text{edge}} = \frac{1}{n} \sum_{i=1}^n |\text{Lap}(\varepsilon_r) - \text{Lap}(\hat{\varepsilon}_r)|, \quad (24)$$

where $\text{Lap}(\cdot)$ represents the Laplacian operator used to extract edges.

The perceptual loss ($L_{\text{perceptual}}$)³⁷ evaluates the similarity between images using high-level features extracted by the pre-trained VGG16 neural network, rather than directly comparing pixel values, which can better reflect the semantic content of images. $L_{\text{perceptual}}$ value is calculated as the L_1 value of the feature tensors from the ground-truth image and the reconstructed image, which is formulated as follows:

$$L_{\text{perceptual}} = \frac{1}{n} \sum_{i=1}^n |V(\varepsilon_r) - V(\hat{\varepsilon}_r)|, \quad (25)$$

where $V(\cdot)$ represents the intermediate feature map by the visual geometry group (VGG) network. A smaller value indicates a higher degree of similarity in the high-level semantic content between the two images in the feature space.

IV. EXPERIMENTS

A. Data preparation

The breast microwave imaging dataset is constructed to validate our proposed network. The breast tissues exhibit a wide variation in dielectric properties, approximately ranging from 1 to 60. A Gaussian baseband pulse with a bandwidth of 1 GHz is used as the transmit signal. An antenna array with 18 sources is placed around the breast model, and the received field data are processed through the inverse algorithm BIM incorporating prior contrast constraints and a multi-frequency strategy³ to generate initial breast microwave images. A total of 45 microwave clinical image slices from public datasets³⁸ are collected, which is far from sufficient for training the neural network. Due to the difficulty in acquiring clinical breast microwave images, the data augmentation strategy (axis flipping, Gaussian blurring, multiplicative noise, and additive noise) is adopted. The final result is 3000 images with a size of 256×256 . Both datasets are divided into training, validation, and testing sets in a ratio of 7:2:1.

B. Implementation details

The Adam optimizer is employed to train the network parameters, with a batch size of 8. The initial learning rate is set to

1×10^{-3} to control the step size of parameter update. To achieve fine-tuning of the learning rate during later training stages, a cosine scheduler is adopted to gradually decay the learning rate following a cosine function to a predefined minimum. This effectively balances the need for rapid convergence in the early stages and precise tuning in the later stages. The maximum number of training epochs is set to 100. The convolutional kernel size is set to [3, 3]. All experiments are conducted on an NVIDIA GeForce RTX 4090 GPU with 64 GB of memory.

Furthermore, ablation studies are conducted to evaluate the impact of key components of the proposed method on performance. Several representative networks are also implemented and compared on the microwave imaging reconstruction task, thereby validating the effectiveness of the proposed method and demonstrating its competitiveness compared with existing approaches.

C. Evaluation metrics

The Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and SSIM metrics are introduced to quantitatively evaluate and compare model performance.

The MSE measures pixel-wise reconstruction error, with smaller values indicating better reconstruction quality. It is expressed as

$$MSE = \frac{1}{N} \sum_{i=1}^N (\epsilon_r^i - \hat{\epsilon}_r^i)^2, \quad (26)$$

where ϵ_r^i and $\hat{\epsilon}_r^i$ represent the ground-truth and reconstructed values of i th pixel, respectively, and N is the number of pixels.

The PSNR is used to measure the similarity between images, defined as the ratio of the maximum power of the signal to the power of the noise. A higher value indicates less distortion, and its unit is dB. It is calculated as

$$PSNR = 10 \times \log_{10} \left(\frac{L^2}{MSE} \right), \quad (27)$$

where L is the maximum of the pixel value.

The SSIM is introduced in the loss function of Sec. III, and its specific definition can be found in Eq. (22).

V. RESULTS AND DISCUSSION

A. Quantitative evaluation

The quantitative evaluation results on the breast dataset are presented in Table I. The traditional BIM imaging,³ bicubic interpolation method, and seven representative network architectures are implemented and compared: SRCNN,³⁹ Cycle-Gan,⁴⁰ Pix2pix,⁴¹ UNet,⁴² MPRNet,⁴³ UNet with EfficientNet⁴⁴ backbone (UNet-Eff), and UNet with ResNet50⁴⁵ backbone (UNet-Res50). Among them, UNet-Eff, UNet-Res50, and MPRNet have demonstrated excellent performance in tasks such as blood vessel segmentation and image deblurring. In this study, these network architectures are introduced to the microwave image enhancement task.

TABLE I. Comparison with various models on breast dataset. Bold values indicate the best performance per metric.

Methods	PSNR (dB) ↑	SSIM ↑	MSE ↓	Params (M) ↓	FLOPs (G) ↓
BIM	17.79	0.6096	0.0261	...	...
Bicubic	17.79	0.6363	0.0300	...	...
SRCNN	21.65	0.7482	0.0086	0.07	4.53
Cycle-Gan	18.59	0.8108	0.0184	16.91	71.94
Pix2pix	21.11	0.7856	0.0103	57.18	21.35
UNet	32.34	0.9557	0.0007	31.04	54.75
UNet-Eff	28.34	0.9200	0.0019	12.83	3.15
UNet-Res50	32.73	0.9630	0.0006	63.56	30.94
MPRNet	38.58	0.9767	0.0004	20.13	1707
MIENet (Ours)	39.14	0.9904	0.0002	13.52	21.58

Table I summarizes the performance on the breast dataset using these traditional methods and advanced deep learning approaches, as well as the number of parameters and Floating-Point Operations (FLOPs). The BIM method that solely relies on inverse problem solving or bicubic interpolation method cannot address the ill-posed nature and lack of details in microwave imaging, resulting in poor image quality. The performance of the SRCNN based on deep learning surpasses that of traditional inverse scattering imaging methods. However, the early super-resolution architecture SRCNN, which consists of only three convolutional layers, fails to adequately handle complex reconstruction tasks. The classic Cycle-GAN and Pix2Pix introduce adversarial training with generator and discriminator networks, enabling the model to learn richer image texture features and further improve image reconstruction performance. The UNet series (UNet-Eff and UNet-Res50) achieve notably improved results compared with traditional methods and adversarial training approaches. The encoder of UNet captures multi-scale contextual semantic information through downsampling, and the decoder restores high-resolution spatial details through upsampling. The skip connections directly transfer the underlying features of the encoder to the decoder, effectively preserving details such as edges and textures. However, the PSNR and SSIM values of UNet, UNet-Eff, and UNet-Res50 based on the general convolution module are still far lower than those of MPRNet and the proposed MIENet. MPRNet learns contextual features through two encoder sub-networks and includes a high-resolution branch for outputting the texture of the image. This multi-stage architecture and cross-stage feature fusion strategy enable the neural network to have extremely advanced image restoration performance, but lead to a substantial increase in FLOPs. The proposed MIENet achieves the best performance on breast imaging, with PSNR, SSIM, and MSE reaching 39.14 dB, 0.9904, and 0.0002, respectively. By contrast, the parameters and FLOPs of the proposed MIENet are reduced to 67.16% and 1.3% of those of MPRNet, respectively, greatly lowering resource consumption. Compared with the top-3 network UNet-Res50, the proposed network's parameters and FLOPs are reduced to 21.3% and 69.7%, respectively. These results demonstrate that the proposed MIENet

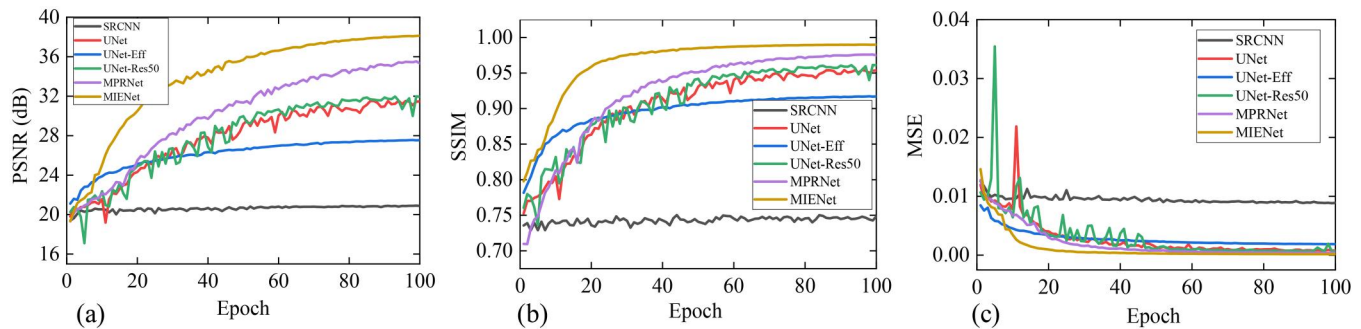


FIG. 4. Evolution curves with epochs of various models on the breast validation dataset. (a) PSNR. (b) SSIM. (c) MSE.

maintains excellent performance while achieving good lightweight characteristics. A detailed analysis of model complexity is provided in Sec. V G. Figure 4 shows the metric variations with epochs for different networks on the breast dataset. Due to the adversarial training nature of these two adversarial networks (Cycle-GAN and Pix2Pix), they cannot be effectively compared with the training curves of other networks. Therefore, the evolutionary curves of these two generative adversarial networks are not included in Fig. 4. It can be observed that the proposed network converges earlier and more stably compared with the other networks. The proposed network consists of the designed GPRM and the classic CBAM mechanism. The GPRM effectively decouples and fuses scattering tissues with different dielectric properties, while the CBAM mechanism focuses attention on strong scattering regions that are prone to errors. The entire network learning process is guided by a hybrid loss function. This loss function can provide a smoother and more instructive optimization direction for gradient descent, effectively alleviating oscillations and accelerating convergence to improve reconstruction quality.

B. Visual comparison

Figure 5 shows the reconstructed images and corresponding error maps for the breast dataset, respectively, using various deep

learning architectures. The error maps are computed as the absolute difference between the enhanced images and the ground-truth images, where brighter regions indicate larger errors. The first row of each figure displays the reconstructed images annotated with their PSNR and SSIM values. These results illustrate the reconstruction performance of different networks on the breast dataset. The UNet, UNet-Res50, UNet-Eff, MPRNet, and the proposed MIENet all successfully enhance the fundamental structure of the breast tissue, with MIENet achieving the highest PSNR and SSIM values. Figure 6 provides zoomed-in details of several randomly selected images from the breast dataset. Each row represents a different image, and each column shows the reconstructed result from a different architecture. It can be seen that MIENet enhances the fine details more accurately than other networks.

C. Ablation study

To investigate the contribution of each key component module (GPRM, CBAM, and hybrid loss) to the network's performance, eight ablation experiments are conducted on the breast dataset. As shown in Table II, the UNet is adopted as baseline (Expt. A), and the key components are gradually incorporated. Expts. A–H represent ablation studies in which the GPRM, CBAM, and hybrid loss modules are either included or excluded,

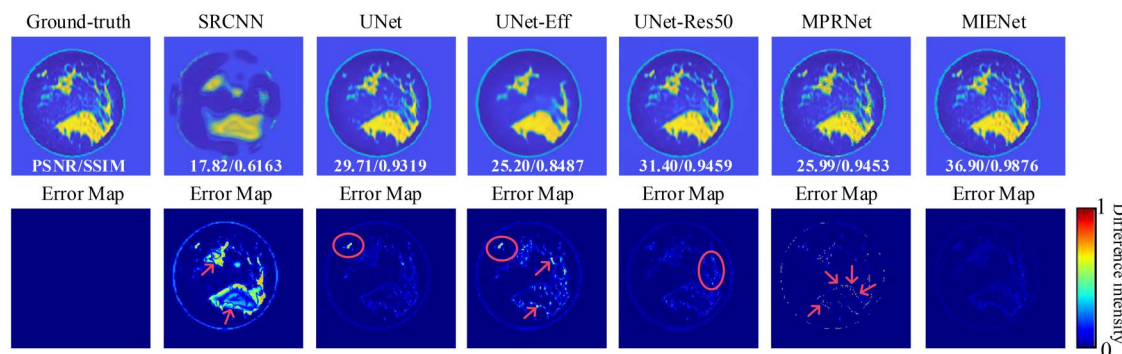


FIG. 5. The image reconstruction results of different networks on breast dataset. Red circles and arrows highlight regions with significant errors.

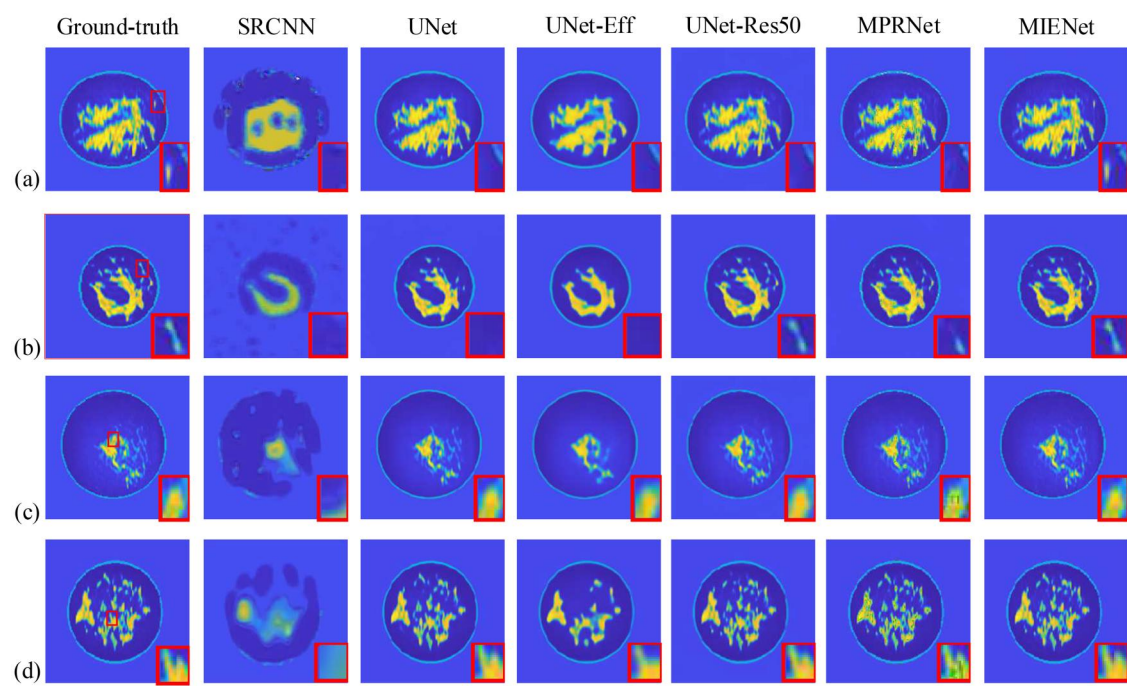


FIG. 6. The visualization results of different networks for the randomly images on breast dataset.

where “√” indicates that the module is used and “—” indicates that it is not used. The hyperparameters and experimental environment employed in these ablation experiments are kept identical. The reconstructed results and corresponding error maps for randomly selected images from the breast dataset in the ablation experiments are presented in Fig. 7.

1. Effectiveness of GPRM

Compared with Expt. A, the incorporation of the GPRM module (Expt. B) on breast dataset leads to a significant

improvement, increasing PSNR by 4.3 dB, SSIM by 0.0248 and reducing MSE by 0.0004. Similarly, Expt. G (without GPRM) achieves PSNR and SSIM values of 37.63 dB and 0.9876, respectively. After integrating the GPRM module (Expt. H), these metrics improve to 39.14 dB and 0.9904, respectively. These results demonstrate the critical contribution of the GPRM to microwave image reconstruction. Breast tissues exhibit complex spatial correlations and large differences in tissue scattering characteristics. The GPRM performs a high-dimensional decomposition of the initial field based on multi-group convolution and feature clustering using few-group convolution. The residual connections in this module also provide a stable reference framework for the image reconstruction process. Therefore, the model incorporating GPRM can better reconstruct different tissue regions, thereby reducing artifacts and detail loss during reconstruction.

TABLE II. Ablation experiments of the proposed MIENet architecture on breast dataset. Bold values indicate the best performance per metric.

EXPTs	GPRM	CBAM	Hybrid loss	PSNR (dB)↑	SSIM↑	MSE↓
A (baseline)	...	...	...	32.34	0.9557	0.0007
B	√	...	...	36.65	0.9805	0.0003
C	...	√	...	37.12	0.9845	0.0003
D	...	...	√	36.73	0.9825	0.0003
E	√	√	...	37.18	0.9833	0.0002
F	√	...	√	39.11	0.9900	0.0002
G	...	√	√	37.63	0.9876	0.0003
H	√	√	√	39.14	0.9904	0.0002

2. Effectiveness of CBAM

In the experiments without and with CBAM (Expt. A → Expt. C), the PSNR, SSIM, and MSE are improved from 32.34 dB, 0.9557, and 0.0007 to 37.12 dB, 0.9845, and 0.0003, respectively. These results confirm that the addition of the CBAM effectively contributes to a significant increase in both PSNR and SSIM and a corresponding reduction in MSE. Strong scattering areas (fibroglandular regions in the breast) are usually the most prone to reconstruction errors. The CBAM automatically assigns higher weights to strong scattering regions through channel and spatial sub-attention modules, actively highlighting key feature

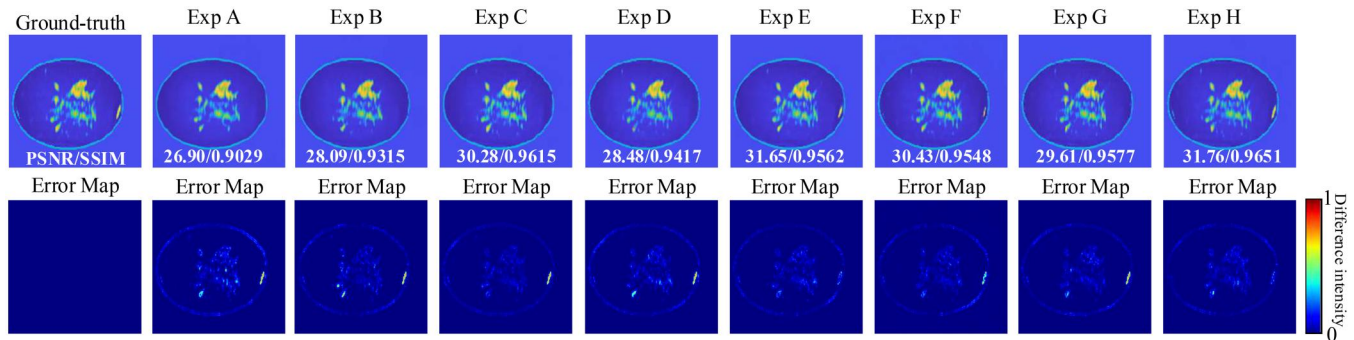


FIG. 7. Ablation experiment results on breast dataset.

channels and spatial regions related to strong scattering tissues. Therefore, the model incorporating the CBAM helps us to reduce image reconstruction errors.

3. Effectiveness of hybrid loss

In the experiments without and with the hybrid loss (Expt. E $\rightarrow$ Expt. H), the PSNR and SSIM improve from 37.18 dB and 0.9833 to 39.14 dB and 0.9904, respectively, while the MSE remains at 0.0002. It can be observed that the hybrid loss significantly improves PSNR and SSIM, but the reduction in MSE is less pronounced. This can be attributed to the fact that the hybrid loss comprehensively evaluates and optimizes various aspects of the image (such as pixel values and structural features), rather than focusing solely on minimizing the pixel-wise MSE metric. The proposed hybrid loss function integrates multiple dimensions of constraints, including structural continuity, boundary clarity, and structural consistency. This ensures that the reconstruction results are physically reasonable and visually realistic.

Figure 8 plots the evolution curves with epochs for the ablation experiments on the breast validation dataset. The curve for Expt. H remains stable and converges quickly, confirming that each module contributes positively to the performance metrics.

With the collaborative effect of each module, the proposed method converges the fastest and most stably, as shown by the dark brown line in Fig. 8.

D. Impact of various parameters on the performance of the proposed model

To comprehensively understand the impact of various model parameters on performance, this section compares the effects of key model parameters, including learning rate, minimum learning rate scheduling value, kernel size, and number of network layers on model performance.

1. Learning rate

It is a core hyperparameter in deep learning training. It controls the step size for updating model weights and determines the magnitude of network parameter adjustment in each iteration. Table III compares the model performance under different learning rates. It can be observed that the model performs poorly at a learning rate of 1×10^{-2} . This is because an excessively large learning rate causes overly aggressive updates, which easily miss the global optimum solution. The learning rate of 1×10^{-3} provides a moderate step size, allowing the model to converge stably and

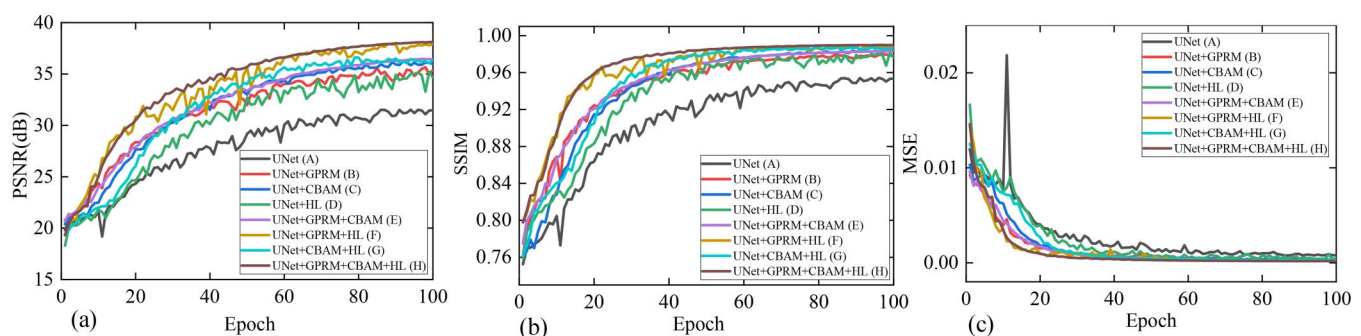


FIG. 8. Evolution curves with epochs in ablation experiments. (a) PSNR. (b) SSIM. (c) MSE.

TABLE III. Comparison of model performance with different learning rates.

Learning rate	1×10^{-2}	1×10^{-3}	1×10^{-4}	1×10^{-5}
PSNR(dB)↑	34.07	39.14	38.18	34.69
SSIM↑	0.9749	0.9904	0.9906	0.9818
MSE↓	0.0005	0.0002	0.0002	0.0004

achieve the optimal reconstruction performance. When the learning rate further decreases to 1×10^{-4} and 1×10^{-5} , the excessively small parameter update step size weakens the model's global search ability. This leads to the optimization process getting stuck in local optima, resulting in a decline in reconstruction performance.

2. Minimum learning rate scheduling value

It is a constraint used in learning rate scheduling to limit the lower bound of decay during the later stages of training. Table IV compares the model performance under different minimum values of the learning rate scheduler. It can be observed that the model performance is relatively low when this lower bound is set to 1×10^{-3} . This occurs because the imposed lower bound is a bit high, leading to overly large parameter updates in the later training stages. The network oscillates near the optimal solution and cannot converge precisely. When the minimum value is reduced to 1×10^{-4} , the model achieves optimal performance in terms of both PSNR and SSIM, indicating that this minimum value enables fine adjustment of parameters. When the minimum value is further decreased to 1×10^{-5} and 1×10^{-6} , performance varies only slightly while remaining generally favorable. Overall, appropriately reducing the minimum learning rate scheduling value can help the network achieve fine convergence, thereby improving the stability and fidelity of reconstruction.

3. Kernel size

It refers to the spatial dimensions of the filter (i.e., the convolution kernel) used during feature extraction, which determines the size of the local receptive field. As shown in Table V, different convolution kernel sizes all achieve good performance and have relatively little impact on the performance of the proposed model. However, the number of model parameters varies significantly across different kernel sizes. The number of parameters for a convolution kernel of size $[k, k]$ is $\text{paras} = c_{\text{in}} c_{\text{out}} k^2$. Thus, with the same input and output channels, the number of parameters of a

TABLE IV. Comparison of model performance with different minimum learning rate scheduling value.

Minimum learning rate scheduling value	1×10^{-3}	1×10^{-4}	1×10^{-5}	1×10^{-6}
PSNR(dB)↑	38.14	39.14	39.09	39.11
SSIM↑	0.9896	0.9904	0.9900	0.9904
MSE↓	0.0002	0.0002	0.0002	0.0002

TABLE V. Comparison of model performance with different kernel sizes.

Kernel size	[1, 1]	[3, 3]	[5, 5]	[7, 7]
PSNR(dB)↑	39.06	39.14	39.49	38.98
SSIM↑	0.9900	0.9904	0.9909	0.9899
MSE↓	0.0002	0.0002	0.0002	0.0002

[5, 5] kernel is approximately 2.78 times that of a [3, 3] kernel. The proposed model adopts a [3, 3] convolution kernel, which effectively balances the receptive field size and the number of parameters. This is also a common setting in the field.

4. Number of network layers

It represents the depth of the model and determines the feature extraction capability. A larger depth leads to a more complex model. Table VI compares the model performance under different numbers of network layers. As the number of network layers gradually increases from 2 to 4, the PSNR and SSIM metrics continuously improve. This indicates that a shallow network with only 2 layers is insufficient to fit complex microwave breast imaging data, thus exhibiting underfitting. When the network deepens to 4 and 5 layers, the model performs well and this depth is sufficient to fit the training data. However, as the number of channels doubles with each additional layer, the total number of model parameters will increase sharply. There is a large difference in the parameter count between the 4- and 5-layer networks, which are 13.52 and 54.06 M, respectively. When the number of layers is further increased to 6, performance drops significantly. The underlying reason is that the model with a huge number of parameters learns many irrelevant features, resulting in overfitting and reduced generalization ability. Therefore, considering both generalization ability and computational cost, a 4-layer network structure can achieve the optimal balance.

E. Physical justification for network design

This paper designs a Group Progressive Residual Module (GPRM). The BIM input already encodes a physics-modeled approximation of the scattering scene. For complex non-uniform media, the initial BIM image can be regarded as a nonlinear coupling of the scattering contributions from multiple targets. Accordingly, the initial multi-group convolution is designed to perform a high-dimensional decomposition of this initial field estimate, which helps retain and extract clearer mode information.

TABLE VI. Comparison of model performance with different number of network layers.

Number of network layers	2	3	4	5	6
PSNR(dB)↑	30.72	36.94	39.14	39.61	37.30
SSIM↑	0.9454	0.9844	0.9904	0.9912	0.9864
MSE↓	0.0013	0.0003	0.0002	0.0001	0.0003

Subsequently, the few-group convolution module facilitates information exchange and fusion among groups, thereby clustering the refined feature patterns into core patterns that represent the corresponding physical tissues. A residual connection is integrated between the input and output of this module, providing a stable reference framework.

To intuitively understand the GPRM's mechanism, we extracted internal feature maps from the GPRM. The proposed network is configured with 16 multi-group convolution modules and 4 few-group convolution modules. Figure 9 displays the feature maps of the multi-group [Figs. 9(b1) and 9(b2)] and few-group stages [Figs. 9(c1) and 9(c2)] during the group progressive convolution process. The multi-group stage generates a larger number of feature maps to express different patterns. Then, in the few-group convolution process, it can be observed that the upper left of Figs. 9(c1) and 9(c2) primarily represent fibroglandular tissue, while the upper right of Figs. 9(c1) and 9(c2) primarily corresponds to the circular skin tissue. The two images at the bottom of Figs. 9(c1) and 9(c2) correspond to the dielectric images of the global tissue. This demonstrates that the GPRM effectively decouples and fuses scattering tissues with different dielectric properties.

The application of the CBAM is motivated by the physical observation that regions with strong scatterers (such as fibroglandular and skin tissue) are prone to significant errors in traditional imaging algorithms due to strong nonlinearity. The channel attention sub-module of CBAM first identifies critical feature patterns by aggregating spatial information via global average pooling to produce channel-wise statistics. A small network then generates weights to emphasize channels encoding strong scattering signatures. Subsequently, the spatial attention sub-module identifies the most important spatial locations. A single 2D spatial feature map

is generated by applying average and max pooling along the channel dimension, which captures the aggregate response at each spatial location. A convolutional layer then processes this map to produce a spatial attention mask, assigning higher weights to regions with high activation—typically corresponding to strong scatterers while automatically suppressing channels with weak responses. Thus, during forward propagation, the CBAM actively emphasizes both the critical feature channels and the salient spatial regions associated with strong scatterers. This guided feature refinement enables the network to concentrate its representational capacity on areas where traditional methods are most prone to reconstruction errors due to nonlinear effects.

To visually demonstrate CBAM's effect, the feature maps are extracted and overlaid on the label maps, as shown in Figs. 9(d1) and 9(d2). It is evident that the network effectively focuses on areas with strong scattering (the dark red areas in the heatmaps), which coincide with key permittivity regions in the ground-truth images. This confirms that CBAM can automatically allocate high attention weights to physically critical regions like fibroglandular tissues, thereby ensuring their accurate reconstruction in the final image.

In addition to designing modules for the characteristics of microwave scattering, designing a loss function suitable for high-fidelity microwave imaging is also crucial for network training. This study constructs a hybrid loss function constraint from multiple dimensions, including structural continuity, boundary clarity, and structural consistency, to ensure the physical rationality and visual fidelity of the reconstruction results. Edge loss encourages the reconstructed image to be consistent with the real image at the edges (corresponding to sudden changes in dielectric constant), while perceptual loss utilizes the high-level features of the

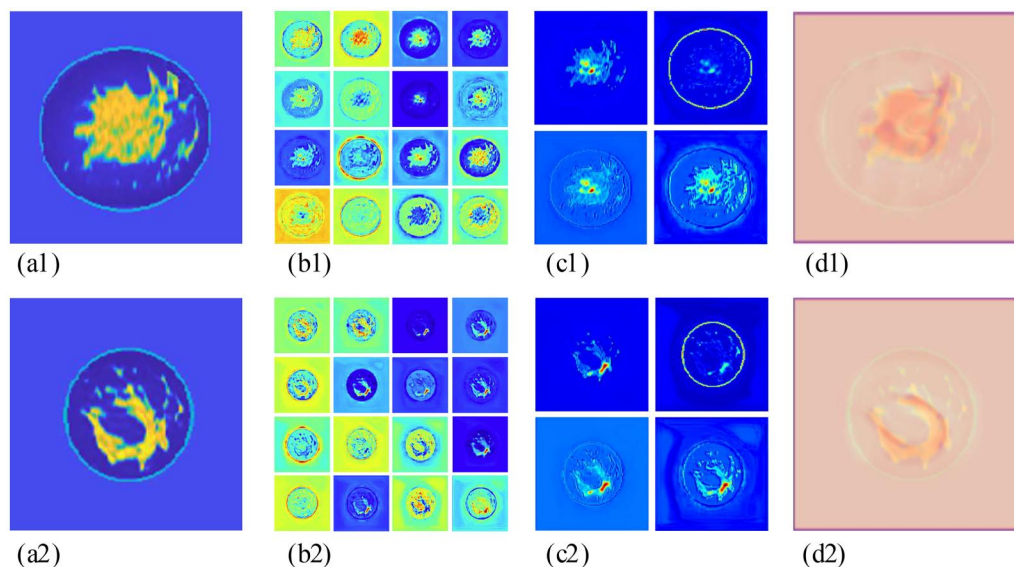


FIG. 9. Internal feature maps of GPRM and CBAM. (a1) and (a2) Ground-truth images. (b1) and (b2) Feature maps of the multi-group stages of GPRM. (c1) and (c2) Feature maps of the few-group stages of GPRM. (d1) and (d2) Feature heatmaps of the CBAM.

pre-trained network to constrain the tissue type of dielectric distribution (as the same type having similar dielectric distribution).

F. Noise analysis

The robustness of the proposed model is tested under different signal-to-noise ratio (SNR) conditions. Various levels of multiplicative noise are added to the original microwave image dataset. Figure 10 shows the performance comparison of several advanced networks on the breast dataset at SNR levels of 10, 15, 20, 30, and 40 dB. Due to SRCNN's relatively low performance, it is not included in the comparison. The curves for UNet, UNet-Res50, UNet-Eff, and proposed MIENet are relatively stable as the SNR decreases from 40 to 10 dB. At high SNR (40 dB), both MPRNet and the proposed MIENet achieve excellent PSNR values. However, MPRNet exhibits a more significant performance degradation at SNR from 40 to 10 dB. Overall, the proposed method maintains stable performance across different SNR levels. This is attributed to the fact that the proposed network structure has the characteristics of high performance and lightweight nature. The proposed GPRM and the classic CBAM module can effectively focus on key scattering regions and suppress noise regions to some extent. Moreover, the lightweight structure is less prone to overfitting on noise. Therefore, the proposed method still achieves excellent and stable performance even under low SNR conditions.

G. Model complexity

The main component in the proposed MIENet is GPRM, and the primary computational cost within the module comes from convolutional operations. Assuming the input and output channel numbers are c_{in} and c_{out} , respectively, and that the channel number has already been converted in the earlier part of the GPRM, the channel number inside the module is c_{out} . With a kernel size of $[k, k]$ and a feature map size of $[f_h, f_w]$, the computational complexity of the two grouped convolutions (with group numbers of $g_{1,m}$ and $g_{2,n}$) is $O((k^2 c_{out}^2 f_h f_w)/g_{1,m} + (k^2 c_{out}^2 f_h f_w)/g_{2,n})$. Thus, the computational complexity of this module can be written

as

$$O_{pro} = O(k^2 c_{out}^2 f_h f_w / g_{min}), \quad (28)$$

$$g_{min} = \min(g_{1,m}, g_{2,n}). \quad (29)$$

Similarly, the complexity of its parameter count is written as

$$O_{pro} = O(k^2 c_{out}^2 / g_{min}). \quad (30)$$

The doubleconv module is usually adopted in the typical UNet, with the computational complexity of $O(k^2 c_{in} c_{out} f_h f_w)$ and the parameter count of $O(k^2 c_{in} c_{out})$. In this study, the number of groups is set to 16 and 4, respectively, so the complexity of the GPRM is approximately one-fourth of that of the original module. Note that the bias parameter, BN layer parameter, and CBAM module are all ignored in the theoretical derivation of this section. The results are generally consistent with the trend and order of magnitude of theoretical derivation. In the breast imaging test, MIENet with the GPRM achieves a reduction in FLOPs and network parameters by 60.58% and 56.44%, respectively, compared with the typical UNet (Fig. 11). The GPRM of the proposed method achieves superior reconstruction performance while effectively reducing network complexity through progressive grouped convolution operations.

H. Limitations and future work

The proposed MIENet achieves excellent performance on public classic breast datasets. However, deep learning-based clinical studies that require large-scale data encounter several practical challenges, including recruiting willing participants, obtaining ethical approval, and securing support from hospitals. Currently, it is difficult to conduct clinical microwave experiments. Thus, the generalization ability of this model in clinical practice still needs to be evaluated and optimized in future work, which is also our subsequent work goal. The primary application of microwave technology in breast imaging is breast density classification and breast

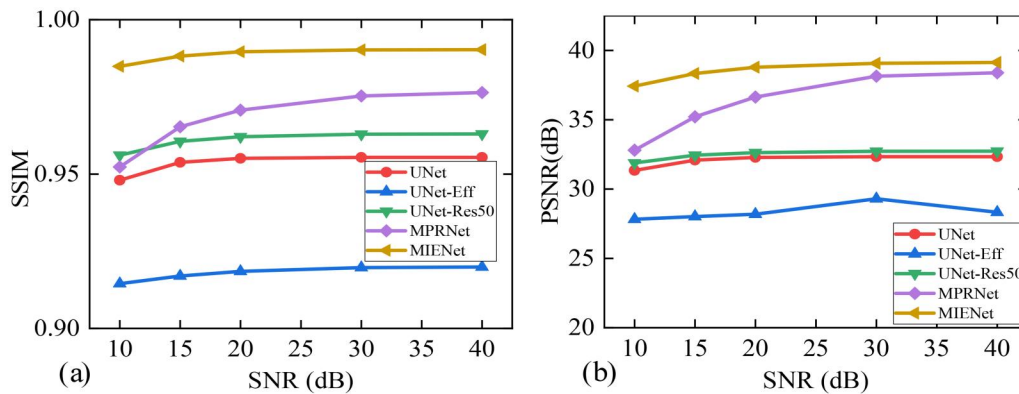


FIG. 10. Reconstruction performance under different noise levels. (a) SSIM. (b) PSNR.

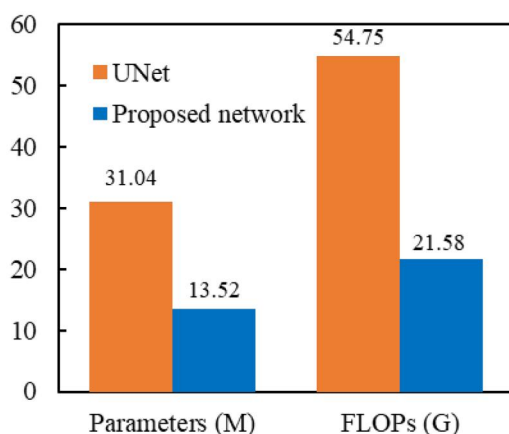


FIG. 11. Model complexity of the proposed network.

tumor detection. Accurate breast density classification can provide crucial guidance in early breast disease screening and personalized diagnosis. We will focus on the practical needs of microwave breast screening and continuously strengthen cooperation with medical institutions to construct a large-scale and multi-center clinical breast microwave dataset. The algorithm in this paper will be integrated and deployed on portable microwave imaging devices. An end-to-end processing flow for microwave signal acquisition and deep learning-based breast image reconstruction will be established. The proposed lightweight and high-fidelity intelligent imaging algorithm can fully take advantage of the green, safe, and miniaturized merits of microwave imaging technology. It is expected to provide a fast, convenient, and highly accessible microwave imaging solution for community healthcare settings.

Furthermore, when creating another image reconstruction application, users need to instantiate the GPRM and CBAM in sequence according to the structure provided in Sec. III B. Then, they can construct a deep learning reconstruction network using the proposed components by calling the hybrid loss function in the training script. Specifically, the proposed GPRM as the core module can be embedded between the encoding and decoding layers of the U-shaped network for feature extraction and detail restoration. CBAM can also be flexibly inserted between the modules of the deep learning network to achieve adaptive weight calibration of the output features and focus on key areas. By calculating the hybrid loss during backpropagation, the corresponding gradients will update the network parameters. All the proposed components can be modularly embedded to form a deep learning network.

VI. CONCLUSION

This study achieves high-fidelity microwave image reconstruction through a physics-assisted deep learning approach. The proposed MIENet is composed of three key components: GPRM, CBAM, and a hybrid loss function, among which the GPRM

component exhibits superior image enhancement effects and lightweight properties. Ablation experiments demonstrate that, under the comprehensive action of these components, the image reconstruction performance is excellent while significantly reducing the resource consumption. Furthermore, the proposed architecture also maintains stable performance under the influence of noise. In the future, the network architecture will be continuously optimized and further lightweighted to suit clinical microwave imaging application scenarios.

ACKNOWLEDGMENTS

This work was supported in part by the National Key Research and Development Program of China (No. 2025YFC2423100), the Natural Science Basic Research Program of Shaanxi (No. 2025JC-YBQN-926), the Key Research Program of Shaanxi Provincial Education Department (No. 25JR133), the Doctoral Research Project of Xi'an University of Technology (No. 103-451123013), and the National Natural Science Foundation of China (NSFC) (Nos. 62104190 and 52205577).

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Min Lu: Conceptualization (lead); Data curation (equal); Funding acquisition (equal); Methodology (equal); Software (equal); Writing – original draft (lead); Writing – review & editing (equal). **Chen Wang:** Conceptualization (equal); Formal analysis (equal); Methodology (equal); Writing – review & editing (equal). **Xia Xiao:** Funding acquisition (equal); Project administration (equal); Writing – review & editing (equal). **Lei Yuan:** Funding acquisition (equal); Validation (equal); Writing – review & editing (equal). **Yuan Yang:** Resources (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of the study are available within the article.

REFERENCES

- Y. Chu, K. Xu, Y. Zhong, X. Ye, T. Zhou, X. Chen, and G. Wang, "Fast microwave through wall imaging method with inhomogeneous background based on Levenberg-Marquardt algorithm," *IEEE Trans. Microwave Theory Tech.* **67**(3), 1138–1147 (2019).
- M. Nakajima and S. Kidera, "A radar-tomographic bidirectional method for quantitative microwave breast imaging," *IEEE Trans. Antennas Propag.* **71**(12), 9814–9829 (2023).
- A. Costa Batista, L. S. Batista, and R. Adriano, "A quadratic programming approach for microwave imaging," *IEEE Trans. Antennas Propag.* **69**(8), 4923–4934 (2021).
- T. M. Habashy, R. W. Groom, and B. R. Spies, "Beyond the Born and Rytov approximations: A nonlinear approach to electromagnetic scattering," *J. Geophys. Res.: Solid Earth* **98**(B2), 1759–1775, <https://doi.org/10.1029/92JB02324> (1993).

- ⁵M. Moghaddam and W. C. Chew, "Study of some practical issues in inversion with the born iterative method using time-domain data," *IEEE Trans. Antennas Propag.* **41**(2), 177–184 (1993).
- ⁶W. C. Chew and Y.-M. Wang, "Reconstruction of two-dimensional permittivity distribution using the distorted born iterative method," *IEEE Trans. Med. Imaging* **9**(2), 218–225 (1990).
- ⁷M. Li, O. Semerci, and A. Abubakar, "A contrast source inversion method in the wavelet domain," *Inverse Probl.* **29**(2), 025015 (2013).
- ⁸X. Chen, "Subspace-based optimization method for solving inverse-scattering problems," *IEEE Trans. Geosci. Remote Sens.* **48**(1), 42–49 (2010).
- ⁹T. Sugitani, S. Kubota, S. Kuroki, K. Sogo, K. Arihiro, M. Okada, T. Kadoya, M. Hide, M. Oda, and T. Kikkawa, "Complex permittivities of breast tumor tissues obtained from cancer surgeries," *Appl. Phys. Lett.* **104**(25), 253702 (2014).
- ¹⁰N. Abdollahi, I. Jeffrey, and J. LoVetri, "Improved tumor detection via quantitative microwave breast imaging using eigenfunction-based prior," *IEEE Trans. Comput. Imaging* **6**, 1194–1202 (2020).
- ¹¹L. M. Neira, B. D. Van Veen, and S. C. Hagness, "High-resolution microwave breast imaging using a 3-D inverse scattering algorithm with a variable-strength spatial prior constraint," *IEEE Trans. Antennas Propag.* **65**(11), 6002–6014 (2017).
- ¹²H. Sato and S. Kidera, "ROI limited unknowns reduction-based contrast source inversion for microwave breast imaging," *IEEE Antennas Wireless Propag. Lett.* **19**(12), 2285–2289 (2020).
- ¹³X. Chen, *Computational Methods for Electromagnetic Inverse Scattering* (John Wiley & Sons, 2018).
- ¹⁴Z. Yin, Y. Jiang, J. Lai, L. Yue, G. Wu, P. Liu, J. Fang, H. Li, J. Dai, and R. Chen, "Cable fault location with convolutional neural network and attention mechanism," *J. Appl. Phys.* **138**(1), 014903 (2025).
- ¹⁵C. Wei, X. Fan, H. Xiao, Y. Kang, X. Huang, C. Li, N. Li, X. Fan, C. Weng, and S. Liu, "Hybrid ResNet-UNet method for multi-parameter acoustic hologram," *J. Appl. Phys.* **138**, 233102 (2025).
- ¹⁶N. E. Khalifa, M. Loey, and S. Mirjalili, "A comprehensive survey of recent trends in deep learning for digital images augmentation," *Artif. Intell. Rev.* **55**, 1–27 (2022).
- ¹⁷M. Ambrosio, S. Franceschini, V. Pascasio, and F. Baseline, "An end-to-end deep learning approach for quantitative microwave breast imaging in real-time applications," *Bioengineering* **9**(11), 651 (2022).
- ¹⁸Y. Qin, P. Ran, T. Rodet, and D. Lesselier, "Breast imaging by convolutional neural networks from joint microwave and ultrasonic data," *IEEE Trans. Antennas Propag.* **70**(8), 6265–6276 (2022).
- ¹⁹H. M. Yao, L. Jiang, and W. E. I. Sha, "Enhanced deep learning approach based on the deep convolutional encoder-decoder architecture for electromagnetic inverse scattering problems," *IEEE Antennas Wireless Propag. Lett.* **19**(7), 1211–1215 (2020).
- ²⁰K. Narendra, B. Martin, C. Gilmore, and I. Jeffrey, "Autoencoder-augmented machine-learning-based uncertainty quantification for electromagnetic imaging," *IEEE Trans. Antennas Propag.* **72**(1), 664–673 (2024).
- ²¹B. Martin, C. Gilmore, and I. Jeffrey, "A long short-term memory approach to incorporating multifrequency data into deep-learning-based microwave imaging," *IEEE Trans. Antennas Propag.* **72**(9), 7184–7193 (2024).
- ²²J. Zhang, C. Li, W. Jiang, Z. Wang, L. Zhang, and X. Wang, "Deep-learning-enabled microwave-induced thermoacoustic tomography based on sparse data for breast cancer detection," *IEEE Trans. Antennas Propag.* **70**(8), 6336–6348 (2022).
- ²³H. Zhou, Y. Cheng, H. Zheng, Q. Liu, and Y. Wang, "Deep unfolding contrast source inversion for strong scatterers via generative adversarial mechanism," *IEEE Trans. Microwave Theory Tech.* **70**(11), 4966–4979 (2022).
- ²⁴Y. Sanghvi, Y. Kalepu, and U. K. Khankhoje, "Embedding deep learning in inverse scattering problems," *IEEE Trans. Comput. Imaging* **6**, 46–56 (2020).
- ²⁵Z. Wei and X. Chen, "Deep-learning schemes for full-wave nonlinear inverse scattering problems," *IEEE Trans. Geosci. Remote Sens.* **57**(4), 1849–1860 (2019).
- ²⁶L. Li, L. G. Wang, F. L. Teixeira, C. Liu, A. Nehorai, and T. J. Cui, "DeepNIS: Deep neural network for nonlinear electromagnetic inverse scattering," *IEEE Trans. Antennas Propag.* **67**(3), 1819–1825 (2019).
- ²⁷P. Shah and M. Moghaddam, "Super resolution for microwave imaging: A deep learning approach" in *2017 IEEE International Symposium on Antennas and Propagation & USNC/URSI National Radio Science Meeting* (IEEE, 2017), pp. 849–850.
- ²⁸V. Khoshdel, M. Asefi, A. Ashraf, and J. LoVetri, "Full 3D microwave breast imaging using a deep-learning technique," *J. Imaging* **6**(8), 80 (2020).
- ²⁹Y. Huang, R. Song, K. Xu, X. Ye, C. Li, and X. Chen, "Deep learning-based inverse scattering with structural similarity loss functions," *IEEE Sens. J.* **21**(4), 4900–4907 (2021).
- ³⁰Á. Yago Ruiz, M. Nikolic Stevanovic, M. Cavagnaro, and L. Crocco, "A deep learning enhanced inverse scattering framework for microwave imaging of piece-wise homogeneous targets," *Inverse Probl.* **40**(4), 045001 (2024).
- ³¹U. Hirose, P. Zhu, and S. Kidera, "Deep learning enhanced contrast source inversion for microwave breast cancer imaging modality," *IEEE J. Electromagn., RF Microwaves Med. Biol.* **6**(3), 373–379 (2022).
- ³²M. Farook Maricar, A. Zakaria, and N. Qaddoumi, "Attention-UNet for electromagnetic inverse scattering problems in microwave imaging," *IEEE Trans. Microwave Theory Tech.* **73**(3), 1290–1304 (2025).
- ³³V. Khoshdel, A. Ashraf, and J. LoVetri, "Enhancement of multimodal microwave-ultrasound breast imaging using a deep-learning technique," *Sensors* **19**(18), 4050 (2019).
- ³⁴P. Mojabi, V. Khoshdel, and J. Lovetri, "Tissue-type classification with uncertainty quantification of microwave and ultrasound breast imaging: A deep learning approach," *IEEE Access* **8**, 182092–182104 (2020).
- ³⁵L. G. Optimization and others, *Gurobi Optimizer Reference Manual, 2020* (Gurobi Optimization, LLC, 2020), Vol. 12.
- ³⁶A. Aghabiglou and E. M. Eksioğlu, "Projection-based cascaded U-Net model for MR image reconstruction," *Comput. Methods Programs Biomed.* **207**, 106151 (2021).
- ³⁷J. Johnson, A. Alahi, and L. Fei-Fei, "Perceptual losses for real-time style transfer and super-resolution," in *European Conference on Computer Vision* (Springer, 2016), pp. 694–711.
- ³⁸M. J. Burfeindt, T. J. Colgan, R. O. Mays, J. D. Shea, N. Behdad, B. D. Van Veen, and S. C. Hagness, "MRI-derived 3-D-printed breast phantom for microwave breast imaging validation," *IEEE Antennas Wireless Propag. Lett.* **11**, 1610–1613 (2012).
- ³⁹C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.* **38**(2), 295–307 (2016).
- ⁴⁰J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *2017 IEEE International Conference on Computer Vision (ICCV)* (IEEE, Venice, 2017), pp. 2242–2251.
- ⁴¹P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition* (IEEE, 2017), pp. 1125–1134.
- ⁴²O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention* (Springer, 2015), pp. 234–241.
- ⁴³S. W. Zamir, A. Arora, S. Khan, M. Hayat, F. S. Khan, M.-H. Yang, and L. Shao, "Multi-stage progressive image restoration," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (IEEE, 2021), pp. 14821–14831.
- ⁴⁴M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *International Conference on Machine Learning* (PMLR, 2019), pp. 6105–6114.
- ⁴⁵K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition* (IEEE, 2016), pp. 770–778.